



DEVELOPER TOOLING

ALEO SCALE GRANT PROPOSAL · JULY 2026

ARC-20 Conformance Kit

Open-source IARC20 conformance QA for Aleo Leo token programs — helping wallet teams, DeFi integrators, and token issuers verify ARC-20 compliance before mainnet.

\$5,000 USD · 8 weeks · 2 milestones

APPLICANT

Panagiotis Pollis (panagot)

TELEGRAM

@panagot

REPOSITORY

github.com/panagot/arc20-conformance-kit

EMAIL

panagiotispollis@gmail.com

LINKEDIN

linkedin.com/in/panagiotis-pollis-33509831

DEMO (TEMPORARY)

arc20-conformance-kit.vercel.app

SECTION 01

Project Overview

ARC-20 Conformance Kit is an open-source integrator QA tool for Aleo's IARC20 fungible token standard. It answers a single question every wallet team, DEX builder, and token issuer must resolve before mainnet integration:

Does this Leo program actually implement ARC-20?

The problem within Aleo

Leo v4 introduced interfaces and dynamic dispatch so DeFi protocols can call any IARC20 token at runtime (e.g. `IARC20@(token_id)::transfer_from_public(...)`). Provable maintains the reference interface library, but there is no public conformance suite — no shared test vectors, no CI gate, and no visual report layer for integrators to regression-test compliance before shipping.

Why privacy / ZK matters

ARC-20 governs private records, shield/unshield flows, and signer-scoped transfers on Aleo's zero-knowledge network. Integrators must trust that token programs correctly implement private transfer, join/split, and approval semantics — not just public balance updates. This kit validates those privacy-critical requirements and maps them to behavioral test vectors.

Anticipated ecosystem impact

- Faster ARC-20 adoption by reducing integration risk for wallets and DeFi teams
- Shared language (12+ behavioral vectors) for token issuers and integrators
- CI-friendly CLI and GitHub Action gates for Leo project pipelines
- Machine-readable reports wallets and services can ingest programmatically
- Complements ProvableHQ/ARCs — does not duplicate the reference interface library

IARC20

LEO V4

DYNAMIC DISPATCH

INTEGRATOR QA

SECTION 02

Team Background

Core team: Panagiotis Pollis — solo developer and maintainer (github.com/panagot)

Contact: panagiotispollis@gmail.com · Telegram: @panagot · [LinkedIn](#)

Full-stack builder — TypeScript/Node CLI, React workstation UI, API integration, spec authoring, documentation, and ongoing maintenance.

Relevant experience

- Shipped working prototype (v0.3): CLI, API, web workstation, 12 vectors, ABI + program validators
- Demo verified for reviewers: arc20-conformance-kit.vercel.app (temporary)
- MIT-licensed open source with integrator documentation
- Parallel tooling: Zcash Indexer Observatory (committee review); Filecoin MemFOC

SECTION 03

Market Potential & Demand

Target user segments

1. Wallet teams validating token compatibility before listing
2. DeFi / DEX builders using dynamic dispatch for arbitrary ARC-20 tokens
3. Token issuers needing a pre-mainnet compliance checklist
4. RPC operators and indexers screening token program interfaces
5. Security / QA engineers auditing third-party token contracts

Pain points uniquely solved

- No shared conformance fixtures — each integrator reinvents ad-hoc checks
- ABI output from `leo build` is not human-readable for non-Leo engineers
- Deployed program source is hard to compare against IARC20 manually
- No CI gate or badge exists for repos claiming ARC-20 compliance

TAM: All Leo developers building or integrating fungible tokens on Aleo.

SECTION 04

System Design & Specification

COMPONENT	DESCRIPTION
Spec layer	IARC20 mirror + behavioral vectors (extensible JSON schema)
Validation engine	ABI validator + program-source validator (Provable API)
CLI	<code>arc20-check</code> — CI exit codes, JSON + text reports
API	REST endpoints for demo, ABI check, program check, spec browser
Web UI	Overview, ABI Check, Program Check, Report, Spec, Vectors
Deployment	Public website + API (production hosting; demo on Vercel temporarily)

Dependencies: Provable API (integrated), Leo toolchain (user-side), Node.js 20+.

SECTION 05

Funding & Milestones

Total request: \$5,000 USD · 8 weeks

MILESTONE 1 · WEEKS 1–4

Integrator-ready v1

\$2,500

- **Sample Leo programs** — conforming + intentionally non-conforming token examples with copy-paste IARC20 template scaffold
- **GitHub Action CI gate** — automated conformance check on PRs with pass/fail summary comment
- **JSON report schema v1** — standardized machine-readable output for wallet pipelines and CI ingestion
- **Report export** — downloadable JSON conformance reports from the web workstation for audit trails
- **Integrator acceptance checklist** — one-page guide for wallet and DeFi teams pre-listing a token
- **Documentation v2** — updated integrator guide, README screenshots, API usage examples
- **Production website** — deploy public website on dedicated hosting (replacing temporary demo)

MILESTONE 2 · WEEKS 5–8

Distribution & ecosystem tooling

\$2,500

- **npm publish** — `arc20-check` installable globally for Leo developers
- **Batch program checker** — validate multiple testnet/mainnet program IDs from a manifest file (wallet screening workflows)
- **Conformance badge generator** — README/CI badge from last check result for open-source token repos
- **Vector expansion** — MintableToken interface checks + 4 additional edge-case behavioral vectors
- **Spec drift alerts** — warn when IARC20 requirements change between consecutive runs
- **Program edition tracking** — surface deployed program edition mismatches vs local ABI builds
- **Maintenance & handoff** — GitHub + website upkeep, community PR review, final delivery report

Budget breakdown

ITEM	AMOUNT
Engineering	\$4,000
Website hosting	\$200
Documentation & integrator guides	\$500
Integrator feedback & spec review	\$300
Total	\$5,000

Out of scope: DEX development, token launch, wallet UI, block explorer, on-chain execution tests.

SECTION 06

Prototype Already Built

- IARC20 spec mirror + 12 behavioral vectors
- ABI + program-source validators (Provable API integrated)
- CLI with CI exit codes and JSON output
- Web workstation v0.3 with live score, category breakdown, vector heatmap
- Automated smoke tests passing on localhost and demo deployment
- Demo: arc20-conformance-kit.vercel.app · Repo: github.com/panagot/arc20-conformance-kit

SECTION 07

Collaborative Value

Partners who benefit: wallet teams, DeFi protocols, Provable/ARC maintainers, token issuers, RPC operators.

Integration: Complements ProvableHQ/ARCs and Provable API. Does not compete with Provable Explorer or AleoScan. JSON report schema designed for downstream wallet and indexer ingestion.